

Autodesk® Scaleform®

Font and Text Configuration Overview

Scaleform 4.3
()

Scaleform C++ API

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: 2.2
: 2012 6 21

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	Overview)	
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はじめに

のフォント

HTML

FreeType 2

/

Flash® Studio
SWF

Flash

Scaleform
Gfx::Translator

FreeType 2

Flash Studio Scaleform
Flash

Flash Scaleform

Flash
Flash

Flash

(Text Field)

Flash

UI

1.1 Flash のテキストとフォントの基礎

Flash Studio

UI

Flash Studio

Flash

GF 1 -

テキストフィールド

Flash

Flash Studio []



ActionScript

GfX::Translator

2

()

a)

b)

Arial

[] []

SWF

Flash 8

()

()

()

1

SWF

Flash Player

Scaleform

()

()

[...]

文字の埋め込み

1

TextField

textfield

(Scaleform

GFx::FontLib

[

...]

)
[]

1		1 KB	450 K
114	- +	12 KB	480 K
596	-	70 KB	630 K
7,583	-	2,089 KB	3,500 K
18,437	-	5,131 KB	8,000 K

“Arial Unicode MS”

500

150K

フォントで使用するメモリの調整

1. UI

Flash

(

2. Gfx::FontLib

(

)
SWF/GFX

Gfx::FontLib

3.

IME

4. GfxEport

-fc
10 30%

5.3

Gfx::FontLib

IME

4

1:

2:

3:

Scaleform

4:

Scaleform

UI

UI

2

3

読み込みフォントの置き換え

Gfx::FontLib

デバイスフォントエミュレーション

(

)

カスタムアセットの生成 Flash

Player

SWF/GFX

Scaleform

3:

Scaleform

Gfx::FontLib

FreeType 2

Gfx::FontLib

4

Scaleform

3

gfxexport

パート ゲームフォントライブラリの作成

UI

UI

UI

UI

Scaleform

fonts_en.swf fonts_kr.swf

SWF

Flash

Flash

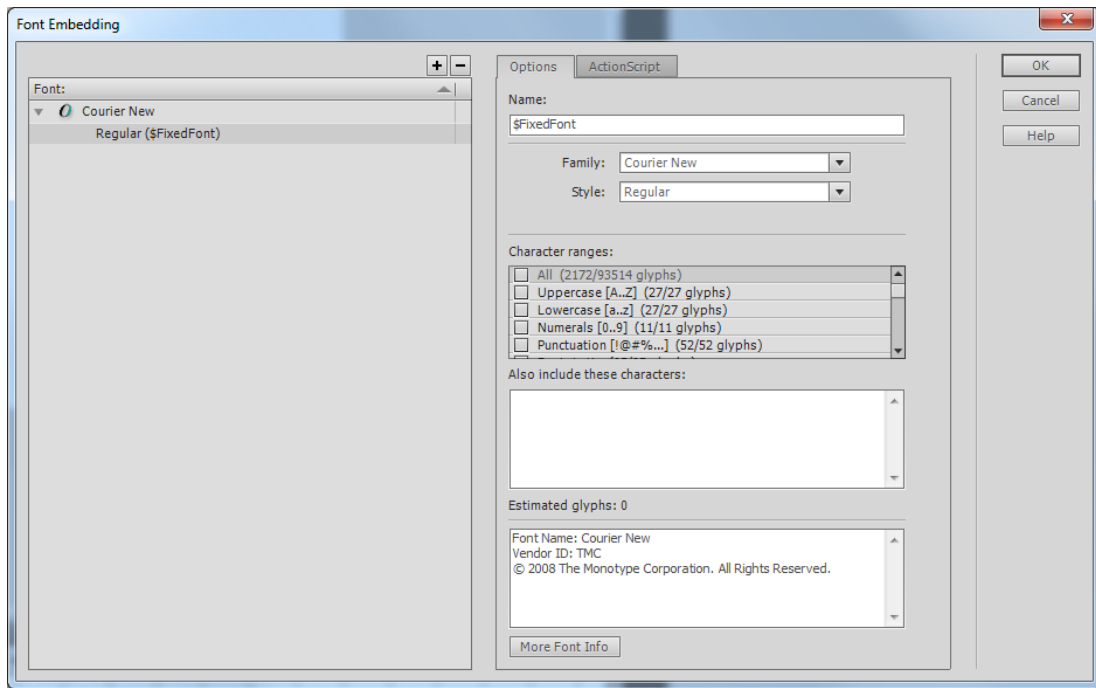
2.1 フォントシンボル

W

Flash Studio

L

[...]



FLA

“\$FixedFont”

“\$ArialGame*”

UI

‘\$’

Web

]

\$

Flash Studio

フォントシンボルの書き出しと読み込み

FLA

/

FLA

FLA

[]

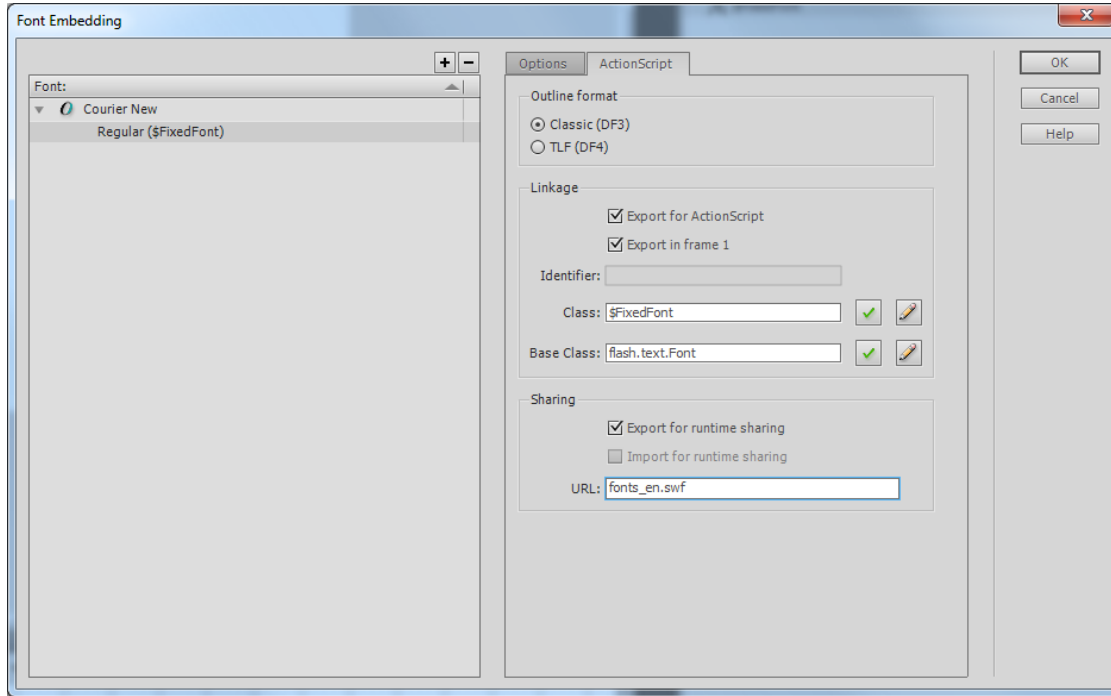
FLA

Flash

/

"Properties"

"ActionScript"



"OK"

Flash

ActionScript

URL

SWF

GfX::FontLib

URL

"fonts_en.swf"

のデフォルトでのフォン

トライブラリがフォントの置換を使用するように設定する必要があります。

```
Loader.SetDefaultFontLibName("fonts_en.swf");
```

GfXPlayer

fonconfig.txt

fontlib "fonts_en.swf"

fontlib

" : \$ "

/

SWF

Scaleform

SWF

```

        TextField.htmlText
        ("Arial"
        )
        ActionScript
    (
    )
    TextFormat

```

書き出されたフォントと文字の埋め込み

```

Flash Studio      CS4
Windows
[Unicode          ] [
243               "English" ] > [
                  11,920
                  "gfxfontlib.swf"

```

```

Flash   CS5
"gfxfontlib.swf"
fonts_en.swf  fonts_jp.swf

```

```

Flash Studio   CS4
"gfxfontlib.swf"
GfX 4.0

```

2.2 gfxfontlib.swf を作成する手順

Flash Studio FLA

1. ' \$TitleFont'
2. ' \$NormalFont'
3. 1 FLA

4.

[]

5.

6.

fonts_<lang>.swf

lang

'en'

'ru'

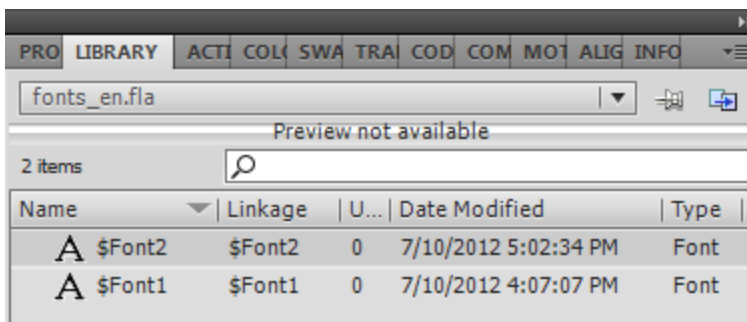
'jp'

'cn'

'kr'

FontMap

FLA



/

パート 多国語化のアプローチの選択

UI

7. 読み込みフォントの置き換え
8. デバイスフォントエミュレーション
9. カスタムアセットの生成

Gfx::FontLib Gfx::FontMap 'fonts_en.swf'

Gfx::Translator

Gfx::Translator Flash Studio

UI UI

3.1 読み込みフォントの置き換え

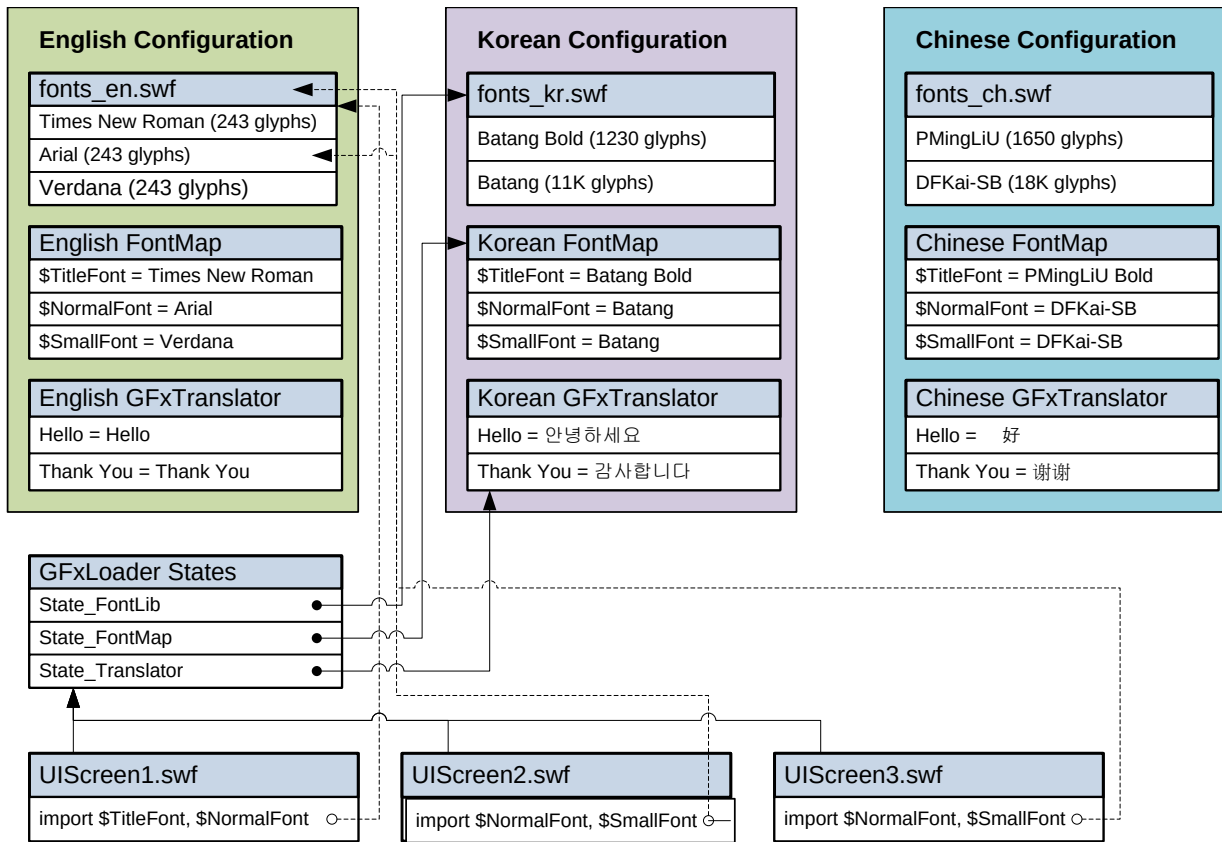
Flash

1 /
 fonts_en.swf

UI

UI UI Adobe Flash Player "fonts_en.swf"

Scaleform



"UIScreen1.swf" "UIScreen3.swf" "fonts_en.swf"

Scaleform
GfX::FontLib 2 GfX::FontMap 1 GfX::Loader 2
Loader::SetDefaultFontLibName()
GfX::FontLib UI

"fonts_kr.swf"

UI

多国語のテキストの設定

		SWF	
	GFX		
GfX::FontMap		GfX::Translator	
Scaleform		"fontconfig.txt"	
1.	1		"fonts_en.swf"
2.			
	GfX::Translator		
3.			GfX::FontMap
4.		"fonts_kr.swf"	
		UI	
1.		GfX::Loader	
	GfX::Translator	Translate	
2.	GfX::FontMap	GfX::Loader	MapFont
		GfX::FontMap	
	GfX::FontMap	3	
3.		(	
	GfX::FontCacheMamanger, GfX::FontProvider	GfX::FontPackParams	)
4.	GfX::FontLib	GfX::Loader	
5.		SetDefaultFontLibName(filename)	
	swf		
	Loader.SetDefaultFontLibName("fonts_en.swf");		
	GfX::FontLib		SWF/GFX
		GfX::FontLib::AddFromFile	
6.	GfX::Loader::CreateMovie		

上での多国語化

"fontconfig.txt" Scaleform Player (Scaleform Player
 SWF/GFX 'fontconfig.txt'
 /fc
 Ctrl+N
 Scaleform Player HUD
 8 ASCII UTF-16

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$TitleFont" = "Times New Roman" Normal
map "$NormalFont" = "Arial " Normal
map "$SmallFont" = "Verdana" Normal
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
tr "Hello" = "안녕하세요"
tr "Thank You" = "감사합니다"
```

```
[FontConfig "Chinese"]
fontlib "fonts_ch.swf"
map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal
tr "Hello" = "你 "
tr "Thank You" = "谢谢"
```

"name"] Scaleform Player HUD [FontConfig

設定ステートメント	意味
fontlib "fontfile"	SWF/GFX GfX::FontLib fontlib
map "\$UIFont" = "PMingLiU"	GfX::FontMap GfX::FontMap UI \$UIFont "fonts_en.swf" UI

tr "Hello" = "你 "	
-------------------	--

Scaleform

Scaleform

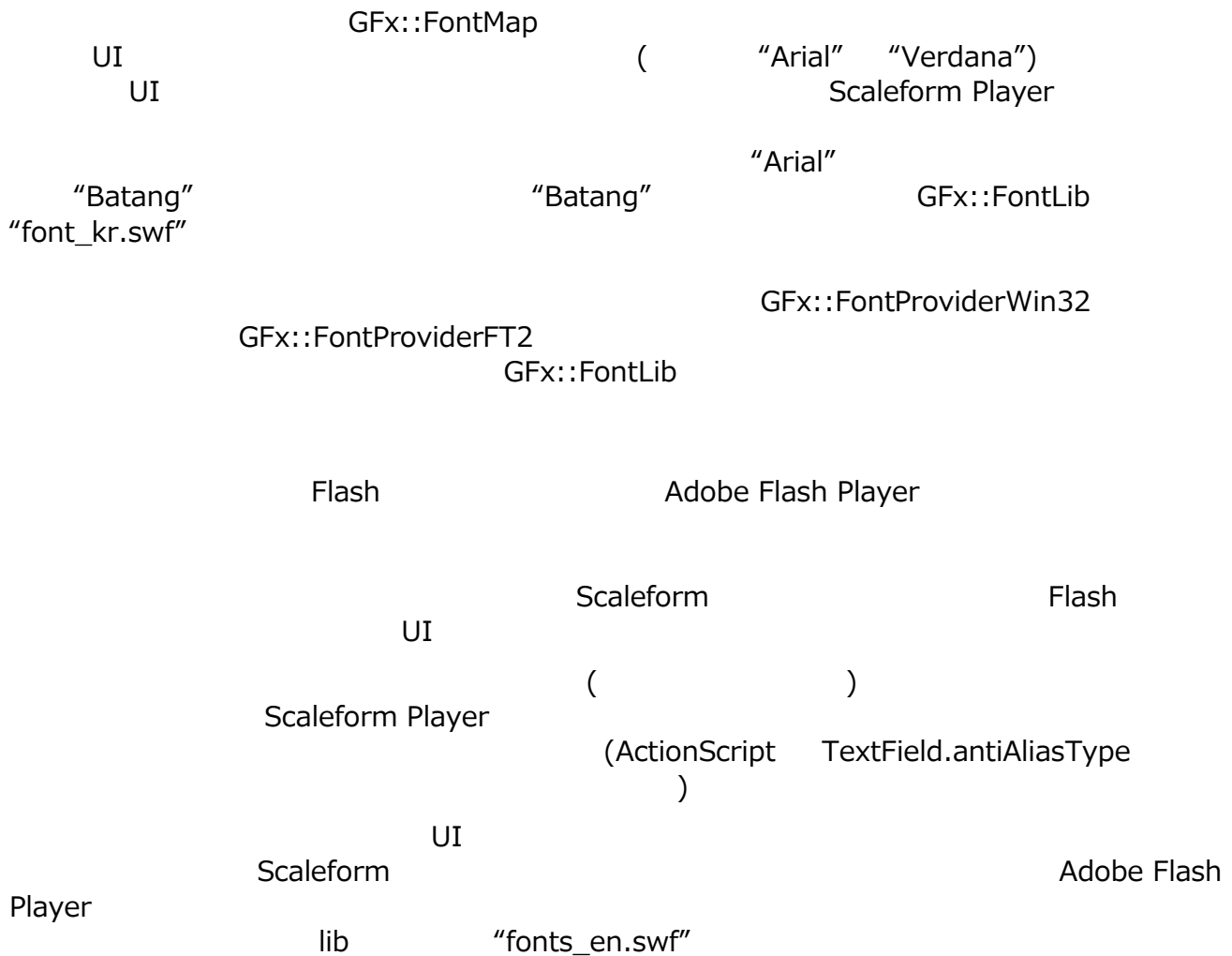
XML

FontConfigParser.h/.cpp

Scaleform

Bin\Data\AS2\Samples\FontConfig and Bin¥Data¥AS3¥Samples¥FontConfig
Scaleform Player "sample.swf"
"fontconfig.txt"

デバイスフォントエミュレーション



ActionScript TextFormat

Flash

```
("$TitleFont"    )
```

("Arial")

"fonts_en.swf"

"fonts_en.swf"

"fonts_en.swf"

UI

フォントライブラリ作成の詳細手順

2

fontconfig.txt

SWF

1.	FLA	ActionScript 2.0	3.0	'main-app.fl'a SWF
----	-----	------------------	-----	-----------------------

2. FLA ActionScript

'fonts_en.swf'

3. 'Library' 'New Font...'

\$Font1	"Family"	"Arial"	"Style"	"Regular"
---------	----------	---------	---------	-----------

4. "Character ranges" 'Basic Latin'

5. 'ActionScript' 'Export for ActionScript' 'Export for frame 1' 'Export for run-time sharing' 'URL' 'fonts_en.swf' OK ActionScript 3.0 Flash Studio 'A definition for this class could not be found...' 'OK'

6. $\frac{\text{'Font2' Family}}{\text{'Style' Family Style}}$

8. 'Library' 'New Font...'
\$Font1 "Family" " " "Style"
"Regular"
9. "Character ranges" 'Korean Hangul
(All)'
10. 'ActionScript' 'Export for ActionScript' 'Export for frame 1' 'Export
for run-time sharing' 'URL' 'fonts_kr.swf'
OK ActionScript 3.0 Flash Studio 'A definition
for this class could not be found...' 'OK'
11. 8 10 '\$Font2' 'Family'
, ,
12. 'fonts_en.swf' 'fonts_kr.swf'
13. main-app.flas
'fonts_en.swf' '\$Font1' '\$Font2'
Ctrl-C 'Copy'
14. 'main-app.flas j Ctrl-V
'Paste' 'Import:'
15. 'main-app.flas 'Classic Text' 'Dynamic Text'
'Family' '\$Font1 *'
'\$TEXT1' ID
16. 2 '\$Font2 *' '\$TEXT2'
17. 'main-app.swf'
18. swf fontconfig.txt Unicode
UTF-8

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = "이것"
tr "$TEXT2" = "은 한국이다!"
```

```

" " " "
"Batang" "BatangChe"
SWF
Flash SWF
fontconfig.txt
gfxexport -fntlst <swfname> *.lst
```

```

19. Unicode UTF-8
GfXPlayer 'main-app.swf' 'This is'
'English!' Ctrl-N fonts_kr.swf fonts_kr.swf
```

3.2 カスタムアセットの生成

```

"UIScreen.swf" "UIScreen.swf"
SWF/GFX
(UI 600K
)
```

UI

UI

(
)

UI

IME

Flash Studio

[

]

(Ctrl + F11)

Scaleform

Flash

[

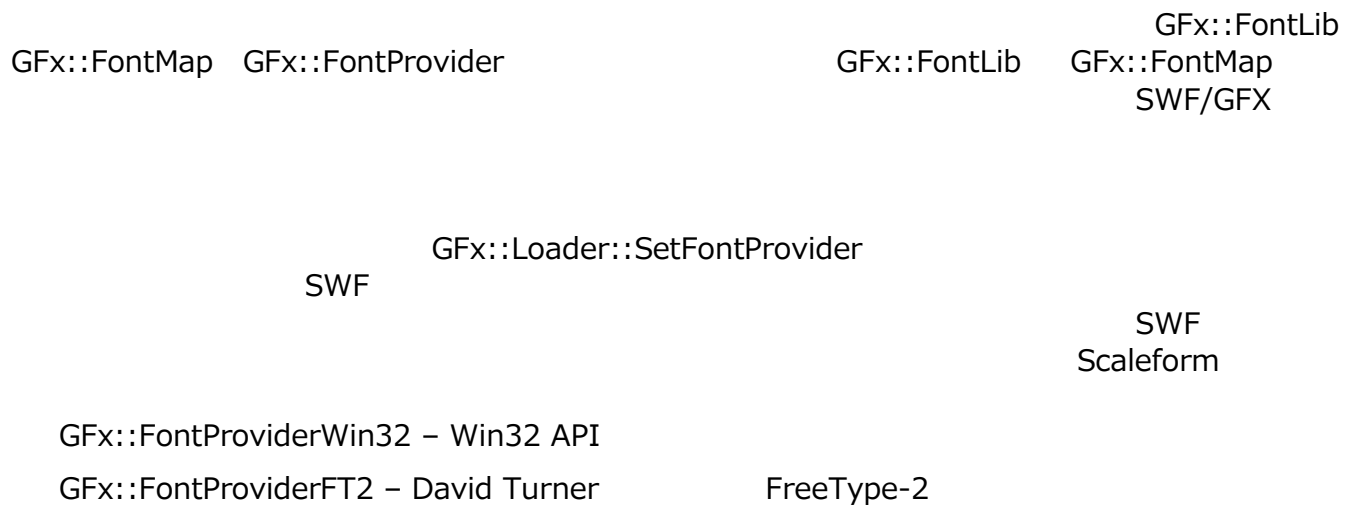
]

パート フォントソースの設定

Flash
HTML

1. ローカルに埋め込まれたフォント
2. SWF/GFX
3. GfX::FontLib SWF/GFX
4. (GfX::FontProviderWin32)

Flash



4.1 フォントの参照順

Scaleform []



4.2 GfX::FontMap

GfX::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

```
( "$TitleFont" )
```

C++

```
#include "GfX/GfX_FontLib.h"
. . .
Ptr<FontMap> pfontMap = *new FontMap;
Loader.SetFontMap(pfontMap);

pfontMap->MapFont("$TitleFont", "Batang", FontMap:: MFF_Bold, scaleFactor = 1.0f);
pfontMap->MapFont("$NormalFont", "Batang", FontMap:: MFF_Normal, scaleFactor = 1.0f);
pfontMap->MapFont("$SmallFont", "Batang", FontMap:: MFF_Normal, scaleFactor = 1.0f);
```

```
"$SmallFont"
3
MapFont
"$NormalFont"
MFF_Original
```

```
scaleFactor
scaleFactor 1.0f
```

GfX::FontMap

GfX::Loader::CreateMovie

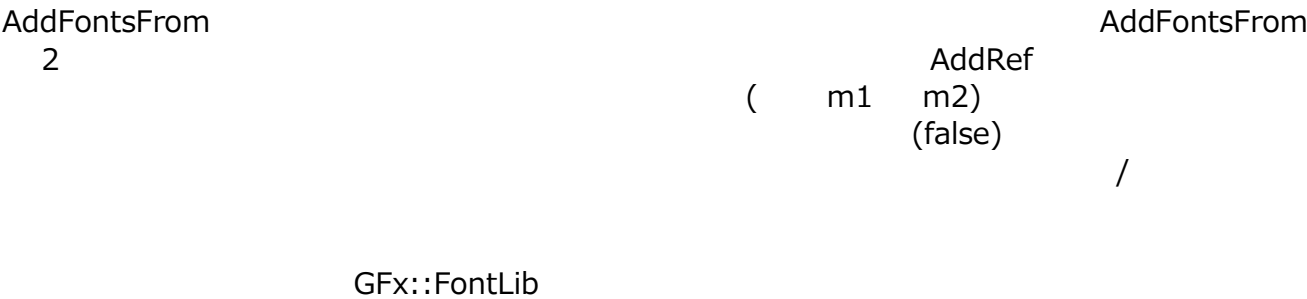
GfX::MovieDef

4.3 Gfx::FontLib



```
#include "Gfx/GFx_FontLib.h"
. . .
Ptr<FontLib> fontLib = *new FontLib;
Loader.SetFontLib(fontLib);
fontLib->SetSubstitute("<default_font_lib_swf_or_gfx_file>");

Ptr<MovieDef> m1 = *Loader.CreateMovie("<swf_or_gfx_file1>");
Ptr<MovieDef> m2 = *Loader.CreateMovie("<swf_or_gfx_file2>");
. . .
fontLib->AddFontsFrom(m1, true);
fontLib->AddFontsFrom(m2, true);
```



4.4 Gfx::FontProviderWin32



```
#include "Gfx/GFx_FontProviderWin32.h"
. . .
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));
Loader.SetFontProvider(fontProvider);
```



“Arial Unicode MS”

abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
𐀀𐀁𐀂𐀃𐀄𐀅𐀆𐀇𐀈𐀉𐀊𐀋𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗𐀘𐀙𐀚
𐀛𐀜𐀝𐀞𐀟𐀠𐀡𐀢𐀣𐀤𐀥𐀦𐀧𐀨𐀩𐀪𐀫𐀬𐀭𐀮𐀯𐀰𐀱𐀲𐀳𐀴𐀵𐀶𐀷𐀸𐀹𐀺

Adobe FLASH

Scaleform

Win32

FreeType

```
vectorRange = Font::DontHint;  
rasterRange = Font::HintCJK;  
maxVectorHintedSize = 24;  
maxRasterHintedSize = 24;
```

ネイティブヒンティングの設定

GFx::FontProviderWin32

CJK

```
fontProvider->SetHintingAllFonts(. . .);
```

or

```
fontProvider->SetHinting(fontName, . . .);
```

SetHintingAllFonts ()

SetHinting()

()

```
void SetHintingAllFonts(Font::NativeHintingRange vectorRange,  
                        Font::NativeHintingRange rasterRange,  
                        unsigned maxVectorHintedSize=24,  
                        unsigned maxRasterHintedSize=24);
```

```
void SetHinting(const char* Name,
               Font::NativeHintingRange vectorRange,
               Font::NativeHintingRange rasterRange,
               unsigned maxVectorHintedSize=24,
               unsigned maxRasterHintedSize=24);
```

Name UTF-8

4.5 Gfx::FontProviderFT2

David Turner FreeType-2

Gfx::FontProviderWin32

FreeType

()

Windows MSVC

```
freetype<ver>.lib - DLL
freetype<ver>_D.lib - DLL
freetype<ver>MT.lib - ( CRT)
freetype<ver>MT_D.lib - ( CRT)

"<ver>" FreeType ( 2.1.9 219)

freetype2/include freetype2/objs
```

Gfx::FontProviderFT2

```
#include "Gfx/Gfx_FontProviderFT2.h"
. . .
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);
```

FontProviderFT2(FT_Library lib=0);

FreeType ()
FreeType FreeType Scaleform

(FT_Done_FreeType)

Gfx::Loader () FreeType

fontProvider->GetFT_Library() FT_Library
FreeType
Gfx::FontProviderFT2 FreeType

フォントのファイルへのマッピング

Win32 API FreeType ()

Gfx::FontProviderFT2

```
void MapFontToFile(const char* fontName, unsigned fontFlags,
                  const char* fileName, unsigned faceIndex=0
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                  Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                  unsigned maxVectorHintedSize=24,
                  unsigned maxRasterHintedSize=24);

    "fontName" ( "Times New Roman")
    "fileName" ( "C:\\WINDOWS\\Fonts\\times.ttf")
    "fontFlags" "Font::FF_Bold" "Font::FF_Italic" "Font::FF_BoldItalic"
                "Font::FF_BoldItalic"
    "Font::FF_Bold | Font::FF_Italic" "faceIndex" FT_New_Face
                                     0
```

Win32 FreeType

フォントのメモリへのマッピング

FreeType

```
void MapFontToMemory(const char* fontName, unsigned fontFlags,
                    const char* fontData, unsigned dataSize, unsigned faceIndex=0
                    Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                    Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                    unsigned maxVectorHintedSize=24,
                    unsigned maxRasterHintedSize=24);
```

```

FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd) {
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
    fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
}

```

()

Gfx::Loader

FreeType

Scaleform

Windows FreeType

```

Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
    "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
    "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
    "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,

```

```
        "C:\\WINDOWS\\Fonts\\verdanaz.ttf");  
    . . .  
    Loader.SetFontProvider(fontProvider);
```

MapFontToFile()

パート フォント レンダリングの設定

Scaleform

$$\begin{pmatrix} 1 & 2 \end{pmatrix}$$

-
-) (mip-map
gfxexport
- (Gfx::FontPackParams
-)

PC Xbox 360 PS3

Wii

Gfx::FontProviderWin32

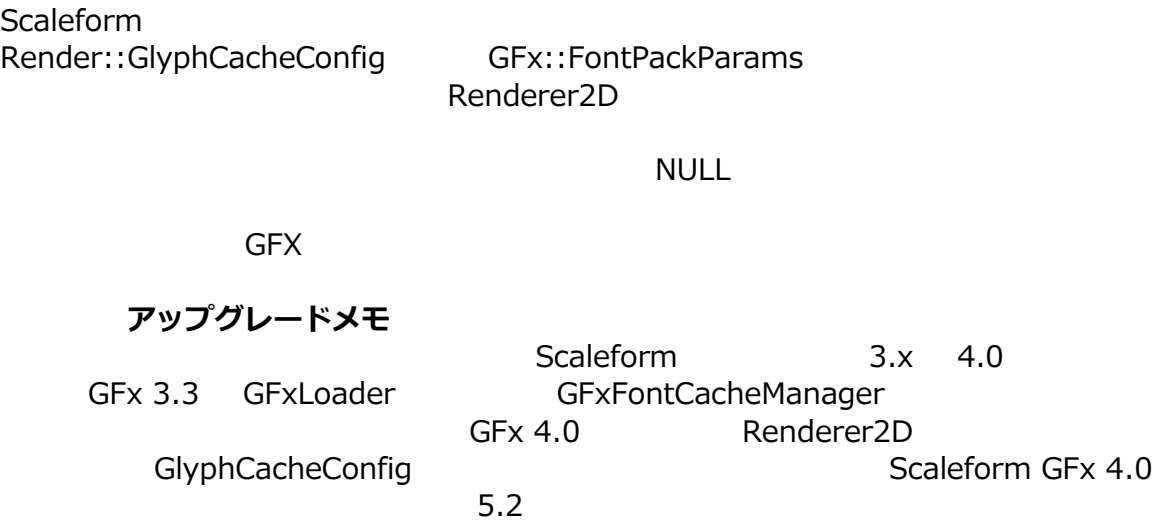
Gfx::FontProviderFT2

```
gfxexport
(PSP PS2 )
```

CPU

Scaleform

5.1 フォントキャッシュマネージャ



GFX::FontPackParams 'gfxexport'

()

```
bool Done = false;

if (Font has Static Textures with Packed Glyphs)
{
    if (GlyphSize <
        FontPackParams.TextureConfig.NominalSize * MaxRasterScale)
    {
        Draw the Glyph as a Texture using Static Cache;
        Done = true;
    }
}
```

```

    }
}
else if (Dynamic Cache is Enabled AND
        GlyphSize < GlyphCacheParams.MaxSlotHeight)
{
    Draw the Glyph as a Texture using Dynamic Cache;
    Done = True;
}

if (Not Done)
{
    Draw the Glyph as Vector Shape;
}

```

5.2 ダイナミックフォントキャッシュの使用

) (10000

```
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

renderer

Render::Renderer2D

HAL

LRU (Least Recently Used)


```

Render::GlyphCacheParams gcparams;
gcparams.TextureWidth    = 1024;
gcparams.TextureHeight   = 1024;
gcparams.MaxNumTextures  = 1;
gcparams.MaxSlotHeight   = 48;
gcparams.SlotPadding      = 2;
gcparams.TexUpdWidth     = 256;
gcparams.TexUpdHeight    = 512;

renderer->GetGlyphCacheConfig()->SetParams(gcparams);

```

TextureWidth, TextureHeight -
2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -
2

TexUpdWidth, TexUpdWidth -
256x512 (128K)) 256x256 128x128

LRU
1MB 4KB 4 K

/ 256 4 K 16

4K 100 256 4K 1 MB

256 2

MaxSlotHeight+2*SlotPadding 2

2-5

(60-80%

(UI 500 2000

1

) () (

Flash

GFX, Dynamic cache	Readability	Animation
<i>Anti-alias for readability</i> <i>Anti-alias for animation</i>		

(())

Scaleform Flash

Scaleform

(O G C Q o g e)

Scaleform

H E F T U V W X Z
Z X V W

1

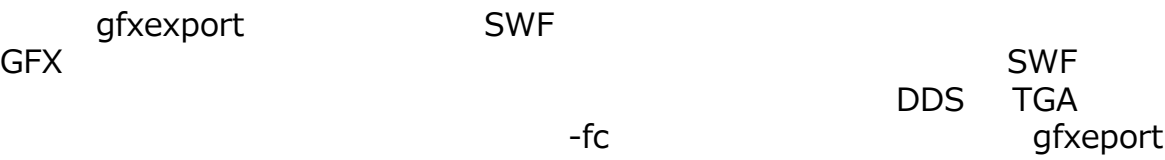
1
Scaleform

[]

Flash

"Zz"

5.3 フォントのコンパクター、gfxexport の使用



コマンドラインのオプション	ビヘイビア
-fc	
-fcl <size>	Flash 256102425625%
-fcm	FontCompactor 1070% 32166412 64123216

5.4 フォント テクスチャの前処理 - gfxexport



普通の x アプリケーションにはエクスポートされたテクスチャの使用は推奨しません。

gfxexport

CPU

Gfx::ImageCreator

Render::Renderer

mip-map ()

'test.swf' 'test.gfx'

gfxexport -fonts -strip_font_shapes test.swf

strip_font_shapes Scaleform

gfxexport

gfxexport

コマンドライン オプション	動作
-fonts	()
-fns <size>	48 () 1 mip-map
-fpp <n>	3
-fts <WxH>	256x256 ('-fts 128' 128x128 1) '-fts 512x128'
-fs	

コマンドライン オプション	動作
- strip_font_shapes	GFX
-fi <format>	TGA8 (grayscaled) TGA24 (grayscaled) TGA32 A8 TGA8 <format> (-i) DDS8 TGA DDS

警告

5.2

5.5 フォント グリフ パッカーの設定

GfxExport

()

Scaleform

```
Ptr<FontPackParams> packParams = *new FontPackParams();
Loader.SetFontPackParams(packParams);
```

```
GfxExport ( .Scaleform )
```

GfxExport

()

```
Loader.GetFontPackParams()->SetUseSeparateTextures(Bool flag);
Loader.GetFontPackParams()->SetGlyphCountLimit(int lim);
Loader.GetFontPackParams()->SetTextureConfig(fontPackConfig);
```

```
SetUseSeparateTextures() "true"
```

"false"

SetGlyphCountLimit()

0 ()

500

)

SetTextureConfig()

```
FontPackParams::TextureConfig fontPackConfig;
fontPackConfig.NominalSize = 48;
fontPackConfig.PadPixels = 3;
fontPackConfig.TextureWidth = 1024;
fontPackConfig.TextureHeight = 1024;
```

NominalSize -
)

(

RAM

"NominalSize"

NominalSize

(

)

PadPixels -

1

TextureWidth TextureHeight -

2

1

.GFX

```
2 Ptr<FontPackParams> packParams = *new FontPackParams();
  Loader.SetFontPackParams(packParams);
  ...
  renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
3 Ptr<FontPackParams> packParams = *new
  FontPackParams(); Loader.SetFontPackParams(packParams);
  Loader.GetFontCacheManager()->EnableDynamicCache(false);
```

()

5.6 ベクトル化の制御

Scaleform

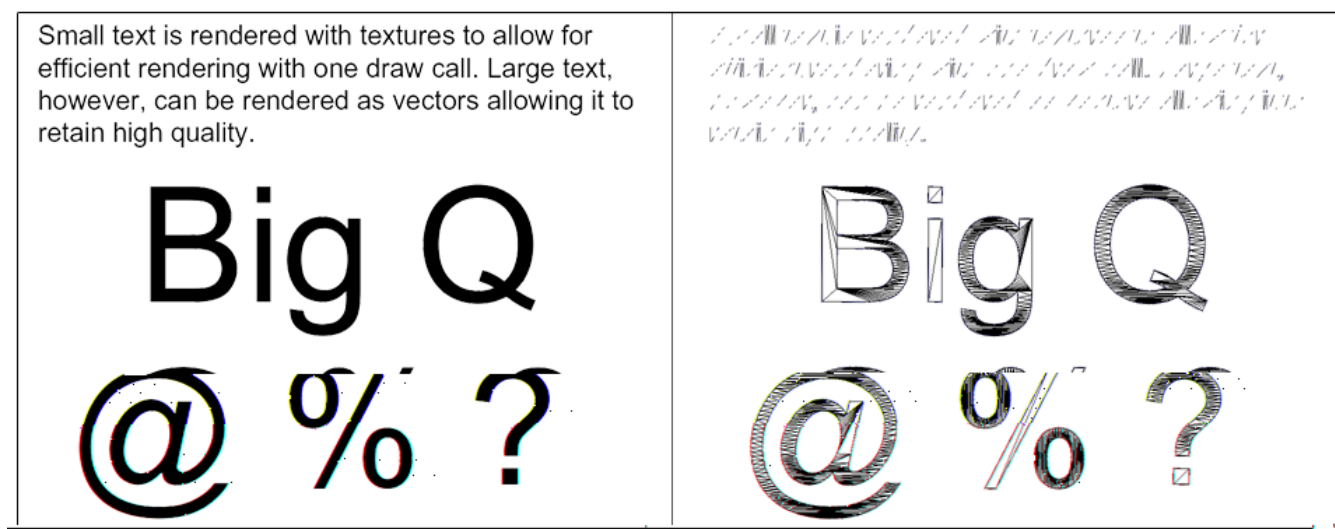
Flash

1

Scaleform

Scaleform Player

Ctrl+W



(MaxSlotHeight)
MaxSlotHeight

Render::GlyphCacheConfig::SetParams

(mip-maps)

GlyphCacheParams:: MaxRasterScale
SetMaxRasterScale

()

MaxRasterScale 1 1.25
1.25

Player
48

60

MaxRasterScale

strip_font_shapes

gfxexport

GFX

パート テキストのフィルタエフェクトと

拡張

Scaleform

Player

Scaleform
Adobe Flash

Flash

Flash

Scaleform

Scaleform

6.1 フィルタの種類、使用可能なオプションと制約

Flash Studio

1



Flash

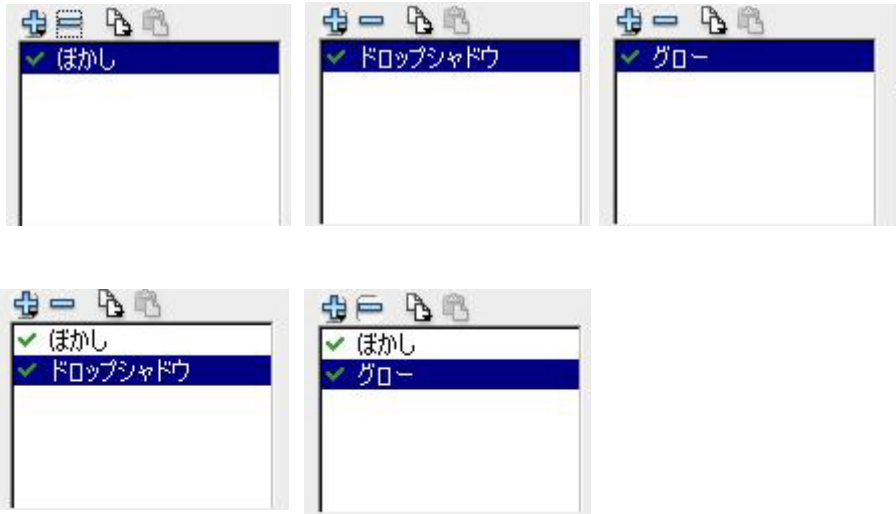
Scaleform

Scaleform
Flash

Scaleform

Flash
Scaleform

-
-
-
-



Flash

$$\left[\begin{pmatrix} \text{Blur X} \\ \text{Blur Y} \end{pmatrix} \right] \left(\begin{pmatrix} \text{Scaleform} \\ 15.9 \end{pmatrix} \right)^2 \left[\begin{pmatrix} \text{Scaleform} \\ 1590\% \end{pmatrix} \right]$$

$$\left[\begin{pmatrix} \text{Flash} \\ \text{Scaleform} \end{pmatrix} \right]$$

Flash GfX
Text *Text*

$$\left[\begin{pmatrix} \text{Flash} \\ \text{GfX} \end{pmatrix} \right] \left(\begin{pmatrix} \text{Flash} \\ \text{GfX} \end{pmatrix} \right)$$

6.2 フィルタ品質

Flash

3x3

2

3x3

[]

3

[]

[]

[]

[]

[]

Flash

Scaleform

Scaleform

[]

[]

:

Scaleform

()

Scaleform

(

	Low quality	Medium quality	High quality
Flash	<i>Text</i>	<i>Text</i>	<i>Text</i>
GFx	<i>Text</i>	<i>Text</i>	<i>Text</i>
	Flash	Scaleform	Scaleform

Smoothin.html

<http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip->

6.3 フィルタのアニメーション

Flash

Scaleform ActionScript

Scaleform

()
()

6.4 *ActionScript* からのフィルターの使用

Scaleform AS2 AS3 /
 DropShadowFilter BlurFilter ColorMatrixFilter BevelFilter
 Flash